

EDITORIAL REPORT

Copy Audit & Rewrite

A complete critique and improvement plan for the homepage, shared interface, navigation, and all 15 interactive exhibits.

SCOPE

homepage, shared exhibit frame, journey/navigation copy, and all 15 live exhibits—including mastheads, guided stories, Run it framing, concept checks, transfer answers, and failure guides.

PREPARED

25 June 2026

Executive summary

ML Lab's core concept and interaction model are strong. The report focuses on making the writing more precise, less repetitive, and more trustworthy without flattening its energy.

15

live exhibits reviewed end to end

12

priority accuracy and trust corrections

5

implementation passes in the rollout plan

ML Lab already has a rare advantage: a recognizable voice and a genuinely differentiated learning structure. The strongest copy is concrete, visual, and active—"drag the point," "watch the line move," "make the model fail." The weakest copy appears when that voice becomes theatrical or absolute. The site often turns a useful teaching simplification into a universal claim.

The main work is not a wholesale rewrite. It is a disciplined edit:

- | | |
|--|---|
| <p>01 Choose one product identity. Use ML Lab as the product, interactive lab as the category, exhibit as the unit, and journey as the guided sequence. Keep atlas, museum, cabinet, jewel, specimen, bench, act, beat, stop, and pass out of core navigation copy. ✓</p> | <p>02 Replace repeated templates with specific outcomes. "Stay twenty minutes..." and "Guardrails off..." appear so often that exhibits begin to sound generated rather than authored.</p> |
| <p>03 Remove unjustified absolutes. Words such as <i>always</i>, <i>never</i>, <i>only</i>, <i>exactly</i>, <i>every</i>, <i>whole field</i>, <i>single most common</i>, and <i>no model can</i> are responsible for most accuracy problems.</p> | <p>04 Separate demonstration-specific behavior from general theory. A surface in this exhibit may become round after standardization; standardization does not guarantee that result in general. A seeded noise dataset may produce $R^2 \approx -0.08$; pure-noise performance does not literally have to equal zero in every finite sample.</p> |
| <p>05 Use validation and test correctly. If a curve is being used to choose degree, λ, capacity, or a threshold, call it validation, not test. Reserve test for the final untouched estimate. ✓</p> | <p>06 Choose one spelling system. The titles use US spellings—<i>Regularization</i>, <i>Generalization</i>—while body copy often uses <i>maths</i>, <i>standardised</i>, <i>minimised</i>. Adopt US English throughout: <i>math</i>, <i>labeled</i>, <i>standardize</i>, <i>optimization</i>, <i>minimize</i>, <i>regularization</i>, <i>generalization</i>, <i>recognize</i>. ✗</p> |

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The report begins with product-level messaging and interface language, then moves exhibit by exhibit through the full learning journey.

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SECTION

Priority corrections before stylistic polish

PRIOR-ITY	LOCATION	CURRENT PROBLEM	REQUIRED CORRECTION
P0	Linear Regression story	"One residual of ten outweighs a hundred residuals of one."	They are equal under squared loss: $10^2 = 100 \times 1^2$. Say "costs as much as."
P0	The Gradient check	A zero gradient is described as meaning "no direction is uphill locally," including at a saddle.	At a saddle, first-order slope is zero but nearby directions can rise and fall. Say only that it is a stationary point.
P0	The Gradient failure guide	A global maximum is associated with a "convex surface."	For maximization, the relevant guarantee is concavity; convexity gives the corresponding guarantee for minimization.
P0	Feature Scaling story/check	Standardization is said to create "equal curvature in every direction" and condition number near 1.	Standardization often improves conditioning but does not remove correlation or guarantee a round bowl.
P0	Train/Test exhibit	The same held-out set is used to choose model complexity and is repeatedly called test data.	Rename it validation data wherever it guides a choice. Keep a final test set untouched.
P0	Classification transfer answer	With 2% defects, the answer says predicting "good" always can score 99%.	It scores 98%. Accuracy alone also cannot reveal the defective-class recall.
P0	Neural Network story	A single unit is said to top out near a coin flip, while the check and demo say about 75%.	Use "about 75% in this XOR setup" consistently.
P0	Data Leakage story/check	Pure-noise out-of-sample R^2 is said to "sit at zero," and a CV-test gap is treated as proof of leakage.	Finite-sample R^2 can be negative; a gap prompts investigation of leakage, shift, variance, or implementation mismatch.
P1	Logistic Regression check	"Any learning rate finds the best line."	With a suitable optimizer and learning rate, training can find the best line in the chosen feature space.
P1	Overfitting story	A degree-12 polynomial with 16 points is said to thread every point exactly.	Say it tracks the training points closely or reaches near-zero training error.
P1	Classification story	Precision is presented as necessarily rising whenever the threshold rises.	Recall is monotone non-increasing; precision often rises but can move irregularly in finite data.
P1	Shared Explain it payoff	Completing checks is said to prove white-board reconstruction and mastery.	Completion is evidence of practice, not proof of mastery. Invite the learner to explain the idea in their own words.

SECTION

Product vocabulary and message architecture

01 Recommended vocabulary

USE	AVOID IN CORE UI
ML Lab	museum / atlas as equal product categories
interactive lab	laboratory, museum, atlas alternating page to page
exhibit	jewel, specimen, object
stage	act, pass, beat used interchangeably
journey	trail, path, map, cabinet used as competing navigation metaphors
experiment	bench when the meaning is not immediately clear
progress	standing, mastery receipts

The visual design can still suggest a museum or collection. The interface should not ask the learner to decode the metaphor before learning the concept.

02 Recommended product promise

Build machine-learning intuition by running the model.

See the idea, change the inputs, push the model until it fails, then explain what happened.

This is clearer and more ownable than simultaneously claiming to be an atlas, laboratory, museum, cabinet, and course alternative.

SECTION

Homepage

01 What works

The headline is active, the product immediately demonstrates that it is interactive, and the four-stage method is a strong differentiator. The guided Foundations journey also gives newcomers a clear route without preventing free exploration.

02 What to change

- The page uses **atlas**, **laboratory**, **exhibit**, **jewel**, **cabinet**, and **museum** as if they were one coherent metaphor.
- “Enter the first exhibit” links to Linear Regression, while the Foundations journey begins with What Is Machine Learning. Either change the destination or name the destination honestly.
- “Form the mental model,” “inspect the implementation,” “learn the operating envelope,” and “prove transfer” sound like internal product language rather than learner language.
- “Now showing · 15 exhibits, all open” spends words on theatrical framing rather than orientation.
- The footer says the product is a museum after the hero says it is a laboratory and the kicker says atlas.

03 Replacement homepage copy

Hero

KICKER

Interactive machine-learning exhibits

HEADLINE

Build intuition by running the model.

BODY

ML Lab turns core machine-learning ideas into hands-on exhibits. See the idea, change the inputs, push the model until it fails, then explain what happened. Explore freely or follow the Foundations journey.

PRIMARY CTA

Start with What Is Machine Learning

Route this CTA to `/exhibits/what-is-ml`.

SECONDARY CTA

Browse all exhibits

If the primary CTA must continue to route to Linear Regression, change its label to **Try Linear Regression**. Do not call it “the first exhibit.”

Collection heading

15 interactive exhibits

Wing copy

WING	REPLACEMENT DESCRIPTION
Groundwork	What learning from data means—and what a model can actually see.
The first models	Predict a number. Draw a boundary.
How models learn	Turn error into an update.
Keeping it honest	Measure what generalizes and spot what misleads.
Going deeper	Combine simple units into flexible models.

How it works

STAGE	REPLACEMENT DESCRIPTION
See it	Build a visual intuition.
Run it	Change the inputs and inspect the model.
Break it	Trigger a failure and diagnose the cause.
Explain it	Apply the idea to a new case.

Foundations description

CURRENT FRAMING IS STRONG BUT COLLOQUIAL. REPLACE IT WITH:

Start with the basic question—what does it mean to learn from data?—then build your first models, learn how they are optimized, and test whether they generalize.

Footer

ML Lab — hands-on exhibits for building machine-learning intuition.

Secondary line:

15 exhibits · designed for laptop and desktop

SECTION

Shared exhibit-frame and microcopy

01 Stage heading

Replace:

Work it in four passes

With:

Work through four stages

02 Stage purposes

STAGE	CURRENT STYLE	REPLACEMENT
See it	Form the mental model	Build the picture
Run it	Inspect the implementation	Try the model
Break it	Learn the operating envelope	Find its limits
Explain it	Prove transfer	Apply the idea

03 Default Run it introduction

Take control of the experiment. Change the scenario, edit the data, and move the controls. Then connect what you see to the equations underneath.

Avoid starting every exhibit with “Guardrails off.” It is lively once and mannered fifteen times.

04 Default Break it introduction

Trigger the failure yourself. Watch the symptom, identify the cause, then repair it.

The existing paragraph repeats the product philosophy instead of telling the learner what to do.

05 Failure guide heading

Replace:

Every way it breaks

With:

Failure modes to recognize

Replace:

Reproduce it live in the Experiment tab

With:

Reproduce it in Run it

The interface no longer uses an Experiment tab.

06 Explain it introduction

Use the live model and the checks below to test whether the idea transfers to a new case.
You can revise an answer after seeing the feedback.

07 Assessment labels

CURRENT	REPLACEMENT
Open	Not answered
Resolved	Correct
Revisit	Try again
Bench cleared	Checks complete
Your standing	Progress
Done — the experiment felt it.	Complete — the experiment registered it.
Commit & reveal model answer	Save my answer and compare

CURRENT	REPLACEMENT
commit your own first — there is no list to recognise	Write your own explanation before comparing.

08 Prediction feedback

CORRECT

That's it. [feedback]

INCORRECT

Not quite. Keep your prediction in mind, then continue to test it. [feedback]

Replace "Now step on and see it happen" with:

Continue to see what happens.

09 Completion payoff

When complete

All {total} checks complete. Before moving on, explain {concept} in your own words: what it does, when it helps, and one way it can fail.

When incomplete

Complete the remaining checks to finish this exhibit. The meter above shows what is left.

Remove "you could now reconstruct this at a whiteboard" and "the mastery model has the receipts." Those lines claim more than the interaction establishes.

10 Journey links

The source links several "Back to the map" actions to `/#map`, while the homepage collection section is `#exhibits`. Use one anchor and one label:

Browse all exhibits →

EXHIBIT 1

What Is Machine Learning?

01 Core critique

The programming inversion is an excellent opening device, but the current copy presents **supervised learning** as the definition of the entire field. “The machine writes the rule” is memorable, yet it hides the human choices of representation, model family, objective, and data. “For seventy years” is unnecessary and dates the copy without improving the idea.

02 Replacement masthead copy

LEDE

In conventional programming, you write the rules. In supervised machine learning, you provide examples with answers, choose a model and objective, and an algorithm fits a rule that can predict new cases.

OUTCOME

You’ll be able to explain what the data supplies, what the learner chooses, and why a model can only be as reliable as the examples and objective behind it.

RUN IT INTRODUCTION

Write a one-feature rule by hand, then train a model on the same labeled examples. Compare what each rule can represent—and how biased examples bend the learned boundary.

03 Story edits

- Replace “Machine learning is the move that gets unstuck” with “Supervised learning handles tasks where examples are available but an explicit rule is hard to write.”
- Replace “let the machine write it” with “fit a rule from labeled examples.”
- Replace “The same move scales to patterns no human could spell out” with “The same approach can fit patterns that are difficult to express as hand-written rules.”
- In “The whole field, in one inversion,” change the heading to **Supervised learning, in one inversion**.
- Add one sentence: “Other forms of machine learning use different signals—for example, unlabeled structure or rewards—but this exhibit begins with supervised learning.”

-
- Replace “the model has no independent notion of truth” with “the training objective rewards agreement with the examples, not agreement with an external ground truth the system was never given.”

04 Check and transfer edits

- Change the first prompt to “What is the essential difference between conventional programming and supervised machine learning?”
- Replace “You never write the rule” with “You do not specify every decision rule directly; you specify the data, representation, model family, and objective from which a rule is fitted.”
- In the résumé-screening transfer, use **a likely cause**, not **the cause**. Biased historical labels are one possibility; sampling, proxy variables, the target definition, threshold choice, or deployment shift can also produce group disparities.

05 Failure-guide edits

- Rename “Biased data, biased rule” to **Biased examples, biased predictions**.
- Do not classify label bias as distribution shift by default. Sampling bias, label bias, target-definition bias, and deployment shift are related but distinct.
- Replace “a straight line needs only a few points; a face recognizer needs millions” with “sample needs depend on noise, dimensionality, model flexibility, and the accuracy required.”

EXHIBIT 2

The Dataset

01 Core critique

The table-to-scatter correspondence is one of the clearest teaching moments on the site. The copy weakens when “the model’s entire world” is repeated as an absolute. A missing variable cannot be used directly, but another feature may act as a proxy; the current check partly acknowledges that while marking an absolute “No” as correct.

02 Replacement masthead copy

LEDE

Every supervised model starts with a table: rows are examples, feature columns are the inputs, and the target is the answer to predict. The model can use only what the table represents, so omissions, errors, and sampling choices become part of the model.

OUTCOME

You’ll learn to read a dataset as a model does—and to spot how one missing variable, bad row, or unrepresentative sample can distort what it learns.

RUN IT INTRODUCTION

Move between the table and scatterplot: each row and point is the same example. Inspect the features and target, then introduce a bad row and watch the fitted trend change.

03 Story edits

- Use “the model’s entire world” once, then switch to literal language.
- Replace “If a column that matters isn’t in the table, the model simply can’t use it” with “The model cannot use an omitted variable directly; it may only recover part of its effect when another recorded feature acts as a proxy.”
- Replace “data work is most of the job” with “data collection and cleaning often determine the practical quality of the result.”
- Replace “the fanciest algorithm faithfully learns the mess” with “a more flexible algorithm may fit the mess more closely rather than correct it.”

04 Check and transfer edits

Correct answer for the missing-location question:

Not directly. The model can use location only if it is represented in the table or partly encoded by another feature that acts as a proxy.

Feedback:

More rows cannot create a variable that was never measured. They can, however, make relationships among the recorded variables easier to estimate.

For the -9999 sensor example, replace “a more powerful model would fit the bad rows even more faithfully” with:

Changing model capacity does not repair a sentinel value that was encoded as a real temperature. Detect and handle the error code before fitting, then reassess whether robust methods are also useful.

05 Failure-guide edits

- Make clear that “one bad row dominates” describes **least-squares fitting in this exhibit**, not every model.
- Replace “no model setting undoes a typo” with “model settings are not the primary repair for a known data error.”

EXHIBIT 3

Regression

01 Core critique

The page makes the target type memorable, but it presents a false binary: regression uses distance and classification uses accuracy. Regression can use several losses and business-defined tolerance metrics; classification can use accuracy, precision, recall, F1, log loss, calibration, utility, or cost. The correct principle is that the metric must reflect the target and the decision.

02 Replacement masthead copy

LEDE

A regression task predicts a quantity on a numeric scale: price, demand, temperature, or time. Because near and far misses are different, you evaluate the size of the errors—not just whether a prediction crossed an arbitrary cutoff.

OUTCOME

You'll be able to recognize a regression task from its target, choose a distance-based metric, and explain what is lost when a continuous outcome is forced into yes/no labels.

RUN IT INTRODUCTION

Predict each score yourself, reveal the observed value, and watch the errors accumulate. Then change the scoring rule to see why the metric must match the decision.

03 Story edits

- Replace “the quality of the whole model is the total of those distances” with “a regression metric aggregates the misses—for example with absolute or squared error.”
- Replace “a guess is simply right or wrong. That’s classification, and accuracy is its score” with “a categorical prediction creates different error types, and the useful metric depends on their costs.”
- Replace the final formulaic contrast with:

Continuous target → regression → predict a quantity → evaluate the size and pattern of the errors. Categorical target → classification → predict a class or probability → evaluate the error types and decision costs.

04 Check and transfer edits

- Replace “accuracy within a tolerance tells you nothing” with “tolerance accuracy is arbitrary unless the tolerance is defined by the real decision before the model is evaluated.”
- A clinically or commercially meaningful tolerance can be a valid secondary metric; it is not a universal default.
- In the house-price transfer, do not conclude that a \$9,000 MAE means the model is “fine.” Compare it with a baseline, the price distribution, segment-level errors, and the use case.
- Replace the answer’s final sentence with:

MAE of about \$9,000 is more interpretable than “3% accurate,” but whether it is good enough depends on the baseline and the decision the estimate supports.

05 Failure-guide edits

Rename the first card:

Accuracy without a justified tolerance

Boundary copy:

A predeclared tolerance can be useful when the application genuinely treats errors inside that band as equivalent. The failure is inventing the band after seeing the predictions or treating it as a universal definition of correctness.

EXHIBIT 4

Linear Regression

01 Core critique

This is the strongest exhibit narratively, but it contains the most visible arithmetic error. The historical opening is elegant yet longer than necessary, and “honest crowd,” “rogue,” “tyranny,” and “appease” moralize observations rather than explain influence.

02 Replacement masthead copy

LEDE

Linear regression fits a straight-line relationship between inputs and a numeric target. Here, “best” means the line that minimizes the sum of squared vertical residuals.

OUTCOME

You’ll see how residuals determine the fit, why squaring makes large misses influential, and when a robust loss may be a better choice.

RUN IT INTRODUCTION

Drag or add points and watch the least-squares line refit instantly. Switch to code to run the same calculation, then compare the result with the visual.

03 Replacement story opening

A straight line can summarize a cloud of points—but only after you decide what counts as a good line. Least squares makes that decision explicit: measure each vertical miss, square it, and choose the line with the smallest total.

Drag a point and watch the fitted line respond. The rest of the exhibit explains exactly why it moves.

This reaches the interaction faster and avoids an unverifiable “most-used statistical tool” claim.

04 Required arithmetic correction

Replace:

One residual of ten outweighs a hundred residuals of one.

With:

One residual of ten contributes the same squared penalty as one hundred residuals of one:
 $10^2 = 100 \times 1^2$.

Replace the related concept-check feedback:

Pleasing one point at residual 20 costs as much as disappointing four hundred points at residual 1.

Do not say it is “worth” more; the penalties are equal before the line refits.

05 Additional story edits

- Replace “honest crowd” with **main cluster** or **bulk of the observations**.
- Replace “the line abandons the crowd” with “the fitted line shifts toward the high-leverage points.”
- Replace “a stray point can wrench the whole line off true” with “an extreme, high-leverage point can pull the fit away from the main trend.”
- Replace “squared error is doing exactly what it was told: make the largest miss small, whatever it costs the rest” with “squared error minimizes the total quadratic penalty, which gives large residuals disproportionate influence.”
- If the points are simulated, remove “real observation.” Use **observed point in this dataset**.

06 Transfer-answer edits

The mansion example currently assumes robust loss is the cleanest fix and says more features or more mansion data cannot help. Replace with:

First decide whether mansions are contamination, a real segment, or evidence that price should be modeled on a log scale. Depending on that diagnosis, useful responses include a robust loss, a transformed target, segment-specific models, or features that explain the price regime. More representative mansion data can improve the model if mansions are part of the population you need to predict.

EXHIBIT 5

Classification

01 Core critique

The threshold interaction is excellent. The copy should distinguish **scores**, **probabilities**, and **decisions**, and it should stop treating accuracy as inherently bad. Accuracy is inadequate when class imbalance or unequal costs make error types matter; it is not meaningless in every classification task.

02 Replacement masthead copy

LEDE

A classifier often produces a score or estimated probability. Turning that score into an action requires a threshold—and moving it changes the balance between false alarms and missed positives.

OUTCOME

You'll read a confusion matrix, distinguish precision from recall, and choose a threshold from the cost of each error—not from a default.

RUN IT INTRODUCTION

Drag the threshold across the score distribution. Watch the confusion matrix and metrics update, then connect each number to the counts it summarizes.

This also fixes the current grammatical fragment: "then the metrics as the maths underneath."

03 Story edits

- Replace "A classifier gives you a probability" with "A classifier gives you a score; some models provide an estimated probability."
- Replace "precision rises" with "precision often rises" when raising the threshold. In a finite sample, precision can move irregularly; recall is the reliably non-increasing quantity.
- Replace "The two move in opposition" with "Changing the threshold usually trades one against the other."
- Replace "Collapse all four into accuracy and you've thrown away exactly the distinctions that matter" with "Accuracy compresses all four into one number and can hide the distinctions that matter."

- Threshold choice depends on error costs, prevalence, probability calibration, operational capacity, and downstream consequences—not costs alone.

04 Check corrections

Cancer-screening answer:

Prioritize high sensitivity/recall while setting an acceptable limit on false positives and confirming the downstream testing pathway can support that operating point.

Do not imply that maximizing recall without constraint is automatically correct.

Replace the $\frac{1}{2}$ -threshold feedback with:

A threshold of 0.5 is natural only when the score is a well-calibrated probability and the two error costs are treated as equal. Many deployments do not satisfy those assumptions.

05 Required arithmetic correction

In the defective-parts transfer, predicting “good” for every part when 2% are defective yields **98% accuracy**, not 99%.

Also replace “recall is likely near zero” with:

Accuracy alone does not reveal recall. Report the confusion matrix, defective-class recall, and precision before deciding whether the model catches defects.

06 Failure-guide edits

- Rename “95% accurate, and useless” to **High accuracy, zero recall**.
- Replace “accuracy is meaningless” with “accuracy alone is inadequate.”
- Separate training interventions—resampling and class weighting—from decision interventions—threshold selection.

EXHIBIT 6

Logistic Regression

01 Core critique

The score → sigmoid → threshold sequence is well chosen. The copy repeatedly calls the output calibrated, although logistic regression probabilities should still be checked for calibration. It also overstates optimization guarantees and implies accuracy is the natural final metric despite the preceding exhibit teaching otherwise.

02 Replacement masthead copy

LEDE

Logistic regression turns a linear score into an estimated probability for a binary outcome. The score stays linear in the features; the sigmoid maps it into 0–1, and a threshold turns it into a decision.

OUTCOME

You'll trace the path from linear score to probability to boundary—and see exactly why a straight boundary fails on a curved problem.

RUN IT INTRODUCTION

Train the model and scrub through the updates. Toggle the probability field, then add a nonlinear feature to compare a straight boundary with a curved one.

03 Story edits

- Replace “a calibrated probability” with **an estimated probability**. Calibration is a property to evaluate, not a guarantee from the model name.
- Replace “At $z = 0$ the sigmoid is exactly $\frac{1}{2}$ —a coin flip” with “At $z = 0$ the model assigns probability 0.5, its point of maximum uncertainty.”
- Scope the straight-line statement: at a 0.5 threshold, a binary logistic model with linear features has a linear boundary.
- Replace “gradient descent finds the line” with “with a suitable optimizer and learning rate, training searches for the line that minimizes log loss.”

- Replace “no straight line could rescue them” with “a linear boundary cannot separate all overlapping or curved cases.”
- Replace “then you can report accuracy at the end” with “then report the metric or set of metrics that matches the decision.”

04 Check edits

Replace:

Any learning rate finds the best line.

With:

More epochs cannot overcome the representational limit. With a suitable optimizer, training can find the best boundary available in the current feature space—but that boundary is still straight.

Replace “a bigger model won’t help” in the temperature transfer with **more training on the same one-feature linear model won’t help**. A larger or differently featured model could help; that is the proposed remedy.

05 Failure-guide edits

- Ensure the code comment and visible scenario agree: the failure guide refers to a live XOR loop while the learner-facing copy uses a parabola.
- Replace “the true boundary” with **the boundary supported by the data-generating process** or simply **the curved pattern**.

EXHIBIT 7

Loss Functions

01 Core critique

"The loss is the judge" is a useful organizing metaphor. It becomes misleading when robust losses are described as producing a "truer" line or throwing away extreme data. A robust loss changes influence; whether that is better depends on how the extremes were generated and what the application values.

02 Replacement masthead copy

LEDE

A loss function turns prediction error into the number a model minimizes. Change that function and you change which mistakes matter most—and therefore which fit counts as best.

OUTCOME

You'll compare squared, absolute, and Huber loss, then choose among them by the cost and plausibility of large errors.

RUN IT INTRODUCTION

Use the same dataset with three losses. Add or move extreme points and watch each fitted line respond, then inspect the penalty curves underneath.

03 Story edits

- Replace "Absolute error charges the miss itself, flat" with "Absolute error grows linearly with the size of the miss."
- Replace "a far point is just one more vote" with "a point's influence grows linearly rather than quadratically."
- Replace "a calmer judge, a truer line" with "a less outlier-sensitive loss, a fit closer to the main trend."
- Replace "squared error's obsession is a feature" with "emphasizing large errors may be appropriate when their cost is genuinely disproportionate."

- MSE is common because it is smooth, has a convenient statistical interpretation under Gaussian noise, and gives a closed-form solution for ordinary least-squares regression. It does **not** give every regression model a closed form.

04 Check edits

- Huber is differentiable at the transition, but “smooth to optimize” should not imply all optimization problems become easy.
- Replace “far points can’t dominate” with “far points have less influence than under squared loss.”
- “On clean data the three lines are nearly identical” is demonstration-specific. Say **on this clean, roughly symmetric dataset**.
- Remove “the choice of loss only bites when something far off is at stake.” Different noise distributions and asymmetric costs can matter without visually obvious outliers.

05 Failure-guide edits

Replace:

A robust loss throws away exactly the data you needed.

With:

A robust loss downweights extreme residuals, which can be harmful when those cases are rare but important rather than contaminated.

Replace “On clean data, prefer squared error” with:

Under roughly Gaussian, homoscedastic noise, squared error is an efficient and convenient choice. Under other noise or cost structures, another loss may still be preferable.

EXHIBIT 8

The Gradient

01 Core critique

The visual concept is strong, but the copy conflates a zero first derivative with “no uphill direction,” and the ascent failure guide reverses convex and concave. These are foundational errors and should be corrected before launch.

02 Replacement masthead copy

LEDE

On a differentiable surface, the gradient collects one partial derivative per variable. It points in the direction of steepest local increase; its negative points toward steepest local decrease.

OUTCOME

You’ll read a gradient’s direction and magnitude, see why it is perpendicular to a level curve at regular points, and learn why a zero gradient does not prove you found the best point.

RUN IT INTRODUCTION

Drag the point across the surface and watch the gradient change. Flip to descent to see a local step, then inspect what happens at peaks, valleys, saddles, and flat regions.

03 Story edits

- Remove “the single most important object in all of optimization.” Use **a central tool in optimization**.
- Replace “its length is exactly the slope” with “its norm is the maximum local rate of increase per unit step.”
- Add the condition to the contour statement: at a smooth point where the gradient is nonzero, the gradient is perpendicular to the local level curve.
- Replace “at a summit (or any flat spot) every direction is level” with “at a stationary point, every first-order directional derivative is zero.”
- Replace “the place with no downhill left” with “a point where first-order information alone gives no descent direction.”

- Explicitly name minima, maxima, and saddles.

04 Required check correction

Current feedback says:

$\nabla f = 0$ means no direction is uphill locally ... equally at a saddle.

Replace with:

$\nabla f = 0$ means the first-order slope is zero in every direction. The point may be a local maximum, local minimum, or saddle; at a saddle, nearby directions still rise and fall even though the gradient itself is zero.

05 Required convexity correction

The failure guide currently says a convex surface guarantees the global answer for an ascent problem.

Replace with:

On a concave objective, any local maximum is global. Equivalently, on a convex objective being minimized, any local minimum is global. The trap requires a non-concave/non-convex landscape.

06 Transfer edits

Different final losses with small gradient norms suggest different stationary regions, but do not prove that each run reached a distinct local minimum. Plateaus, saddle neighborhoods, numerical tolerances, and optimization settings can also matter. Recommend comparing validation performance and diagnostics, not simply keeping the run with the lowest training loss.

EXHIBIT 9

Gradient Descent

01 Core critique

The fog-and-hillside metaphor works. The page becomes less trustworthy when every slow run is diagnosed as a small learning rate and every failed run as divergence from a large one. Step size is one cause among conditioning, scale, stochastic noise, plateaus, saturation, poor initialization, and implementation errors.

02 Replacement masthead copy

LEDE

Gradient descent updates parameters by stepping against the gradient. The learning rate controls the step: too small is slow; too large can oscillate or diverge.

OUTCOME

You'll connect the update rule to the path on the loss surface and diagnose whether a slow or unstable run is caused by step size, conditioning, or a flat region.

RUN IT INTRODUCTION

Play, pause, and scrub the optimization path. Change the learning rate and surface shape, then compare steady convergence, slow progress, oscillation, and divergence.

03 Story edits

- Replace "the engine under nearly everything" with **a core optimizer behind many models.**
- Replace "That is the entire algorithm" with **That is basic batch gradient descent.**
- Remove "the most common way a training run dies" and "the first question asked in any ML team."
- Replace "Undertraining is not a broken model; it is an unfinished walk" with "In this scenario, the model is progressing but the step is too small to finish in a practical number of iterations."
- Add: "A flat loss curve can also signal poor conditioning, vanishing gradients, noisy mini-batches, or a representational limit."
- Keep the critical learning-rate value clearly labeled as specific to the displayed quadratic surface.

04 Check and transfer edits

Replace the slow-training model answer with:

A learning rate that is too small is one plausible cause, but first inspect the gradient norm, feature scaling, conditioning, batch noise, and whether the model has reached a plateau. If the evidence points to step size, increase it carefully and watch for oscillation or divergence.

Replace “the failure mode that haunts every real training run” with **a common optimization failure**.

05 Failure-guide edits

- Do not classify learning-rate divergence as the same thing as exploding gradients. They can interact, but they are distinct diagnoses.
- Gradient clipping can cap a step, but it does not repair poor curvature or choose a stable learning rate by itself.
- Replace “standardize so the bowl is round” with “standardize to improve the scale mismatch; correlation may still leave the surface elongated.”

EXHIBIT 10

Feature Scaling

01 Core critique

This page contains the biggest generalization error in the site: standardization is portrayed as guaranteed whitening. Subtracting each feature's mean and dividing by its standard deviation puts marginal scales on comparable footing, but correlated features can still produce an elongated loss surface.

02 Replacement masthead copy

LEDE

Features measured on very different scales can make gradient-based optimization zig-zag and distance-based models let one variable dominate. Scaling puts features on comparable numeric ranges; it often improves conditioning, but it does not remove correlation.

OUTCOME

You'll see how scale changes the optimization landscape, when standardization helps, and which models barely care about it.

RUN IT INTRODUCTION

Compare raw and standardized features on the same problem. Watch the path, safe learning-rate range, and condition number change—without assuming standardization always makes the surface perfectly round.

03 Story edits

- Replace “the model treats a one-dollar change as the same size as a one-year change” with “the raw numerical scales determine how changes in each feature affect distances and gradients.”
- Replace “The data hasn't changed; their description has” with “The information is the same, but the coordinates and parameterization have changed.”
- Replace “steepness equal in every direction, condition number near one” with:

Standardization often reduces scale-driven anisotropy and raises the range of stable learning rates. Correlation between features can still leave the surface elongated.

- Replace “the one-line fix” with **a common first fix**.
- Tree methods are largely invariant to monotonic rescaling in exact arithmetic, but avoid saying scaling “buys nothing” in every implementation or pipeline.

04 Required check correction

Correct answer:

It puts feature axes on comparable marginal scales, which often improves the condition number. It does not guarantee equal curvature because correlated features can still stretch the bowl.

Feedback:

Centering and scaling remove unit mismatch. Decorrelation or whitening is a separate transformation.

05 k-NN transfer edit

Replace “switching models or collecting more data won’t help” with:

Scaling is the first repair for a Euclidean-distance model, but feature relevance and the distance metric still matter. A model that does not compare raw magnitudes—such as a tree—may behave differently, and more representative data can still improve the estimate after the metric is fixed.

06 Leakage wording

The failure guide correctly notes that full-data standardization is often a small leak. Preserve that nuance everywhere; do not use a dramatic deployment-collapse scenario as though scaling leakage alone necessarily caused it.

EXHIBIT 11

Train, Test & Generalization

01 Core critique

The page teaches the right discipline but uses the word **test** for data that is repeatedly inspected and used to select model complexity. The internal logic already says a final test set should be sealed, so the visible labels should follow that rule.

02 Recommended title

Train, Validate, Test & Generalize

If the shorter title must remain, introduce validation explicitly in the lede.

03 Replacement masthead copy

LEDE

A training score tells you how well a model fits data it has seen. To estimate performance on new data, you need held-out evaluation—and a clear separation between data used to tune the model and data used only for the final check.

OUTCOME

You'll see why one small split can be noisy, how cross-validation supports model selection, and why a final test set must stay untouched until the end.

RUN IT INTRODUCTION

Reshuffle the split and compare the variation in one holdout score with the distribution of cross-validation scores. Use validation or cross-validation to choose; reserve the test set for the final estimate.

04 Story edits

- Replace “the only question that matters” with **the question a generalizing model must answer**.

- A held-out error is not necessarily higher than training error in every split. Say **often higher, but it may be lower by chance on a small or easier holdout**.
- Replace “the model is fixed” in the check. Each split changes both the training sample used to refit the model and the examples used to score it.
- Cross-validation does not guarantee that the estimate “barely moves.” Report the mean and variation across folds or repeats.
- Use CV for model selection and a final untouched test set for the final estimate. Do not tell learners that the CV number is “the number to report instead” without this distinction.
- “Every point is tested exactly once” applies to ordinary k-fold CV, not every CV strategy.

05 Replacement cross-validation paragraph

K-fold cross-validation rotates the validation fold: fit on $k-1$ folds, score on the remaining fold, and repeat. Averaging the fold scores uses the training data more efficiently and reveals how sensitive the estimate is to the split. It reduces dependence on one lucky holdout; it does not remove uncertainty.

06 Transfer edit

A stable production gap does not uniquely identify distribution shift. Replace the model answer with an investigation sequence:

Check whether production examples match the training population, whether features are computed identically, whether labels mean the same thing, whether latency or missingness changed, and whether any leakage inflated validation. Then collect production-like evaluation data and retrain or redesign the pipeline as the diagnosis requires.

EXHIBIT 12

Bias & Variance

01 Core critique

The classical U-shaped story is appropriate for this polynomial demonstration, but the copy presents it as a universal law. More importantly, it asks learners to choose the “sweet spot” from a test curve. That curve is validation data by definition.

02 Replacement masthead copy

LEDE

As model flexibility grows, training error usually falls. Validation error often falls at first, then rises when the model starts fitting noise. Bias and variance name the two pressures behind that trade-off.

OUTCOME

You’ll recognize underfitting and overfitting from train–validation behavior, then choose model capacity without touching the final test set.

RUN IT INTRODUCTION

Change polynomial degree and compare training with validation error. Watch the fitted curve move from too rigid to suitably flexible to overly sensitive.

03 Story edits

- Change every model-selection use of **test error** to **validation error**.
- Use “often U-shaped in this classical setting,” not “the honest score is U-shaped.”
- Replace “no amount of data fixes bias” with “with the model class held fixed, more data does not remove approximation bias.”
- Replace “simple models are all bias; flexible ones are all variance” with “simpler models tend to have more bias and less variance; more flexible models tend to reverse that balance.”
- Replace “more data shrinks variance” with “more data often reduces estimation variance.”
- Treat training epochs as a capacity control only in the context of early stopping or iterative fitting.

04 Check edits

- “More data never flips a flexible model into underfitting” is too absolute. Say the expected variance generally falls while the model class stays the same.
- A train–validation gap suggests overfitting but can also reflect split mismatch, leakage, or distribution differences. Use **strong evidence**, not **proof**.
- Replace “a model that aces validation and flops in the world” with “a model that fits training data but performs poorly on held-out data.”

05 Failure-guide edits

Replace:

Reduce capacity toward the sweet spot the test curve marks.

With:

Use a validation curve or cross-validation to choose capacity, then evaluate the selected model once on the final test set.

EXHIBIT 13

Data Leakage

01 Core critique

This is a compelling exhibit and an important differentiator. Its weakness is diagnostic certainty: a disappointing sealed-set score is treated as proof of leakage, when distribution shift, sampling variability, repeated tuning, label mismatch, and implementation differences can create the same symptom.

02 Replacement masthead copy

LEDE

Data leakage happens when information unavailable at prediction time influences training or evaluation. The result is an optimistic score that can collapse on truly unseen data.

OUTCOME

You'll manufacture a strong score from pure noise, locate the leak in the pipeline, and rebuild the evaluation so every fold is genuinely held out.

RUN IT INTRODUCTION

Compare feature selection performed before cross-validation with selection fitted inside each fold. Watch the apparent R^2 vanish when the pipeline stops peeking.

03 Story edits

- Replace "the single most common way" with **one common way**.
- Replace "Any model's out-of-sample R^2 should sit at zero" with:

With no signal, expected out-of-sample performance is no better than the baseline. In a finite sample, R^2 can land near zero or below it.

- Keep the seeded values clearly labeled as one demonstration, not a law.
- Correct the typo "a fit between the model's checks and a true, untouched holdout" to **a gap between**.

- Replace “the direction never does” with “the expected effect is optimistic bias, although any one finite sample can vary.”

04 Check correction

Replace the “most reliable way to catch a hidden leak” item. A sealed test set is important, but a low score does not diagnose leakage by itself.

NEW PROMPT

Cross-validation looks strong, but the final untouched test score is much lower. What does that result justify?

CORRECT ANSWER

Investigate leakage, distribution shift, repeated tuning, split mismatch, and pipeline differences before drawing a conclusion.

FEEDBACK

The sealed test set reveals that the earlier estimate did not transfer. It does not identify the cause on its own; the pipeline and data-generating process still need to be audited.

05 Scaling-transfer correction

Do not assert that a deployment gap is “the leak’s signature, not distribution shift.” Replace with:

Fitting the scaler before the split is a leak and should be corrected with a pipeline. Re-evaluate after the repair; if a deployment gap remains, investigate distribution shift and implementation differences rather than assuming the scaler was the sole cause.

06 Failure-guide edits

- “Score too good to be true” is a warning, not a diagnosis.
- Preserve the strong prediction-time availability test for future features.
- Distinguish low-impact unsupervised leakage, such as global standardization on a large IID sample, from severe target-aware or temporal leakage—without normalizing either practice.

EXHIBIT 14

Overfitting & Regularization

01 Core critique

The penalty dial is an effective visual. The copy should describe regularization as reducing **effective capacity**, not keeping full capacity untouched. It also needs a consistent spelling and a more accurate account of interpolation and weight decay.

02 Replacement masthead copy

LEDE

Regularization adds a cost for large or complex parameter values. It reduces a model's effective flexibility without necessarily changing its nominal architecture or polynomial degree.

OUTCOME

You'll tune the regularization strength from under-penalized to well-balanced to over-penalized—and see how validation error reveals the useful middle.

RUN IT INTRODUCTION

Move λ across several orders of magnitude. Watch the coefficients shrink, the curve smooth, and training and validation error trade places, then connect the behavior to the ridge penalty.

03 Story edits

- Replace “threads every training dot exactly” with **tracks the training points closely and reaches near-zero training error**.
- Replace “keep the capacity; penalize using it” with “keep the same parameterization while reducing its effective flexibility.”
- L1 can drive **some** coefficients to zero; do not label them automatically useless, especially with correlated features.
- Replace “weight decay is exactly this L2 penalty” with “under common SGD setups, weight decay corresponds closely to L2 regularization; the equivalence is not universal across optimizers.”

-
- Replace “dropout and early stopping are regularizers in disguise” with “dropout and early stopping are other regularization methods.”
 - Remove “almost every modern model is regularized by default.”

04 Check edits

- Replace “keep the model’s full flexibility” with “keep the same feature set and degree while discouraging large coefficients.”
- Replace the claim that removing regularization or increasing model size is always wrong. In the stated scenario, tune λ by validation; in another scenario, either change could be justified.

05 Failure-guide corrections

- The “over-penalized” card is currently tagged with the **overfitting** primitive. It is an **underfitting** failure.
- Replace “the best λ is found by validation, never by intuition” with “domain knowledge can guide the search, but validation or cross-validation should choose among candidate values.”
- Use US spelling consistently: **regularization, penalized, standardized.**

EXHIBIT 15

Neural Network Fundamentals

01 Core critique

The exhibit succeeds by showing the boundary change in real time. Its copy overstates the historical role of nonlinear layers and the universal-approximation result, and it contradicts itself about one-unit XOR performance.

02 Replacement masthead copy

LEDE

A neural network composes weighted sums with nonlinear activations. A single linear unit draws one straight boundary; hidden units and nonlinearities let the model represent more complex shapes.

OUTCOME

You'll watch a small network learn XOR, see why the activation matters, and test how too little or too much capacity changes generalization.

RUN IT INTRODUCTION

Train the network and inspect the boundary and weights as they change. Vary the hidden-unit count to compare underfitting, a workable fit, and overfitting on noisy data.

03 Story edits

- Replace "the fix that launched deep learning" with **a foundational idea behind multilayer neural networks**.
- Replace "the lone neuron tops out around a coin flip" with **in this balanced XOR setup, the best linear boundary reaches about 75%**.
- Replace "bend into any shape at all" with **represent a much wider range of nonlinear boundaries**.
- Replace the universal claim with:

With enough hidden units and a suitable activation, a network can approximate a broad class of continuous functions. That is an expressiveness result: it does not guarantee that training will find the function or that it will generalize.

- Replace “each hidden unit is a logistic-style neuron” with **each hidden unit computes a weighted sum followed by a nonlinearity**; this demo uses tanh.
- Replace “more units and more layers buy more shape” with “more units and layers can increase representational capacity, while also making optimization and generalization harder.”

04 Check edits

- Qualify the 75% answer as **in this displayed XOR setup**.
- Replace “more training widens the gap” with “more training can widen the gap once the network is fitting noise.”
- Replace “training loss hitting zero is the warning, not the goal” with “in this small noisy dataset, near-zero training loss paired with worse validation is the warning.” Zero training loss is not universally evidence of failure.

05 Failure-guide edits

- The activation belongs between linear layers; the output activation depends on the task. Avoid “between every pair” as an unconditional architectural rule.
- Replace “the one ingredient” with **the ingredient that prevents stacked linear maps from collapsing into one linear map**.

SECTION

Recommended implementation order

01 Pass 1 — Accuracy and trust

Fix the P0 items first: squared-error arithmetic, gradient stationary-point language, convex/concave reversal, standardization claims, validation/test labeling, classification arithmetic, XOR contradiction, and leakage diagnostics.

02 Pass 2 — Shared voice

Update the homepage positioning, shared stage descriptions, Break it and Explain it introductions, assessment states, and completion payoff. This removes repetition across all exhibits in one change.

03 Pass 3 — Mastheads

Replace the lede, outcome, and Run it introduction on all 15 exhibit pages. These are the most visible and highest-leverage strings.

04 Pass 4 — Narrative and assessment nuance

Apply the sentence-level edits in each exhibit's narrative, concept check, transfer answer, and failure guide. Search globally for:

- Stay twenty minutes
- Guardrails off
- single most common
- always
- never
- only
- exactly
- whole field
- the model has no notion
- the honest score
- test error where the value is used for selection

Each hit should be reviewed rather than mechanically deleted.

05 Pass 5 — Consistency QA

Check:

- US spelling throughout.
- "Probability" only where calibration or probabilistic interpretation is justified; otherwise use "score" or "estimated probability."
- "Validation" for tuning and selection; "test" for final untouched evaluation.
- Demonstration-specific numbers labeled as such.
- Every answer marked correct remains true without relying on hidden assumptions.
- No navigation copy references an Experiment tab or `#map` anchor that no longer exists.

SECTION

Final recommended voice rule

Be vivid about the interaction and conservative about the claim.

"Drag the threshold and watch recall fall" is vivid and testable. "Accuracy is meaningless" is broader than the evidence. "This seeded example produces $R^2 \approx 0.41$ " is vivid and honest. "A test gap proves leakage" is not. The product will feel more expert—not less energetic—when its certainty matches what the experiment actually demonstrates.